



Common Weight Loss Medications

Choosing the Best Option

Below is a list of medications to consider when patients are requesting help with their weight loss journey. These can easily be discussed during a GCN visit.

Metformin

Evidence supports the weight-loss effects of metformin in adults with obesity and without type 2 diabetes. It should be considered in patients with prediabetes. The magnitude of metformin-induced weight loss is modest but clinically significant, and it is achievable at low cost with an agent that has



proven long-term safety, few serious adverse effects, and well-documented favorable non-glycemic effects. To date, metformin is the only pharmacologic weight-loss intervention to demonstrate long-term effects.

Thus, in the absence of contraindications, metformin should be seriously considered as an off-label initial therapy and as an adjunct to anti-obesity medications approved by the US Food and Drug Administration for the management of obesity.

A meta-analysis that included 21 trials and 1,004 participants analyzed the effect of

metformin on BMI in different populations and found that in patients with obesity, BMI was reduced by 1.3 units. A sub-analysis found that metformin had the most pronounced effect in the population with BMI greater than 35 kg/m².

Evidence regarding the timeframe in which weight loss might be expected to occur is inconsistent. Commonly, however, in trials with longer follow-up periods, weight loss generally starts after 4 weeks of treatment with metformin and occurs mainly during the first 6 to 12 months of continuing metformin therapy.

Safety and Side Effects

When metformin is used and prescribed appropriately, serious adverse events are extremely rare. The most common side effects are gastrointestinal—diarrhea, nausea, flatulence, vomiting, and abdominal discomfort. These are less frequent with postprandial use and with extended-release, rather than immediate-release, formulations. Given the lack of prospective data on the effect of metformin on weight loss, it is unclear whether weight loss is associated with gastrointestinal side effects. Since the magnitude of weight loss during the DPP trial (and its maintenance during the DPPOS study) was directly related to adherence to metformin therapy, such an association seems unlikely.

The main contraindication associated with metformin is severe renal impairment. Here are some guidelines for dosing based on eGFR:

- eGFR less than 45 mL/min/1.73 m²: Refrain from starting metformin.
- eGFR 45 to 60 mL/min/1.73 m²: Prescribe a maximum total daily dose of 1,500 mg (or 1,700 mg if prescribing an immediate-release formulation).
- eGFR 30 to 45 mL/min/1.73 m² and already on metformin therapy: Adjust to a maximum daily dose of 1,000 mg.



Note. From Metformin [Photograph], by My Health Explained, n.d., My Health Explained (<https://www.myhealthexplained.com/diabetes-information/diabetes-articles/benefits-of-metformin-a-wonder-drug>)

Bottom Line

It is well-known that a small but sustained reduction in body weight (3% to 5%) is associated with improved glucose metabolism, blood pressure, and lipids, and is a strong predictor of diabetes prevention. Available evidence supports the use of metformin as an initial and adjuvant weight-loss medication, especially in the presence of prediabetes, severe obesity (BMI ≥ 35), use of antipsychotic drugs, or polycystic ovary syndrome. It should be considered a long-term treatment, particularly in patients who demonstrate a good response. The aim is to achieve a dosage of 1,500 mg/day or more (or adjusted by renal function), leveraging extended-release formulations and slow titration.

For more information, please visit this study from the [Cleveland Clinic Journal of Medicine](#).

Phentermine

Phentermine is one of the most-prescribed weight loss medications. If a patient is receiving it and they lose weight on it, they should maintain on it unless the side effects are a problem.



Note. From Phentermine [Photograph], by Well Rx Pharmacy, n.d., Well Rx Pharmacy (<https://wellrxpharma.com/product/phentermine-37-5mg/>)

Phentermine monotherapy has been shown to produce about 7.5% weight loss long-term at several year follow-up. 15mg phentermine plus topiramate 100mg has been very well-studied and produces an average of 10% bodyweight loss, though about a third of patients will lose 15% total bodyweight.

Phentermine is a tool. It's great for people with slow basal metabolism who don't have CAD, uncontrolled HTN, uncontrolled glaucoma, or uncontrolled anxiety.

Please be sure to review CURES if prescribing. Also, please be sure to prescribe to the patient's preferred pharmacy, as it can't be transferred after it is signed in HealthConnect.

Please don't use fenfluramine (Fen-Phen), as it has distinct cardiac side effects and is only indicated for the treatment of seizures. Some members will ask for it because they used fen phen in the past, so be careful.

Naltrexone/Bupropion (Contrave)

Naltrexone/bupropion (Contrave) combines an opioid receptor antagonist (naltrexone) with a dopamine and norepinephrine reuptake inhibitor (bupropion) in an extended-release tablet. It is labeled as an adjunct to increased physical activity and a reduced-calorie diet for chronic weight management in adults who have a body mass index (BMI) of at least 30 kg per m² or who have a BMI of at least 27 kg per m² with one or more weight-related comorbidities, such as type 2 diabetes mellitus, hyperlipidemia, or hypertension.

Naltrexone/bupropion produces clinically significant weight loss when combined with a diet and exercise program. It decreases body weight in patients with diabetes, but its effect on diabetic outcomes is not known. A significant proportion of patients will experience adverse effects. It should not be prescribed for patients with preexisting heart disease until the effects are known. If prescribed, naltrexone/bupropion should be used with a diet and exercise program and should be discontinued if at least a 5% weight loss is not achieved within three months.

Here at the VMC, we don't prescribe Contrave, but rather prescribe the individual components of bupropion and naltrexone. Bupropion SR is usually started at 150mg qAM, with naltrexone 50mg ½ tab daily.

For more information, please visit [this study from the American Family Physician](#).



Note. From Naltrexone [Photograph], by Accord, n.d., Accord (https://www.accordhealthcare.us/product/naltrexone-hydrochloride-tablets/?variation_id=1505)

Topiramate

Alongside its anti-epileptic properties, topiramate has been shown to cause weight loss in some patients. The exact way that it does so is not fully understood. However, the off-label use of topiramate has been studied in several clinical trials. In multiple tests, topiramate produced more significant weight loss in individuals who took the medication than those in the placebo group.



Some possible mechanisms include appetite suppression (reduced calorie intake), preventing the body from storing excess fat, and lowering some fat and cholesterol levels. It might also affect the brain's reward system, potentially preventing binge eating.

Topiramate may influence a hormone called leptin, which plays a role in fat storage. Some studies have found that topiramate can lower leptin levels, which may contribute to weight loss. Additionally, topiramate may affect genes

related to neuropeptide Y, which can influence the body's weight regulation system. Moreover, topiramate may work by blocking a substance called carbonic anhydrase that affects how the body metabolizes sugar and lipids (fat cells).

Overall, topiramate has various effects on the body's metabolism and weight management. It can lead to weight loss, lower blood glucose levels, and improve cholesterol and triglyceride levels.

Topiramate can be combined with phentermine (Qsymia) for added benefit.

For more information, please [check out this article from Sesame Care](#).

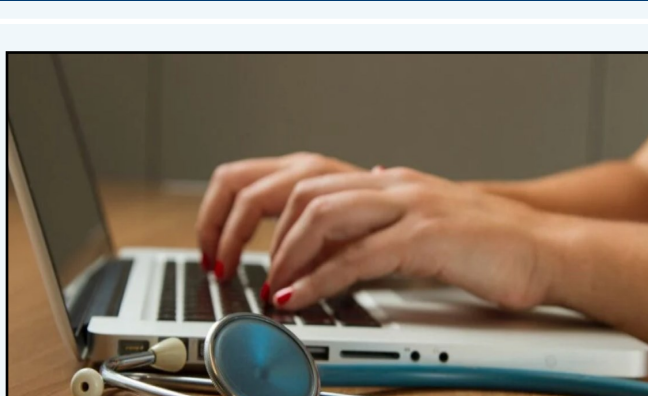
Drug Monitoring

Phentermine: Monitor blood pressure after starting, then every 3-6 months.

Topiramate: Check electrolytes and creatinine initially and with dose adjustments, then every 6-12 months. Monthly pregnancy tests are recommended while taking topiramate for weight loss.

Naltrexone: Check LFTs at baseline, 3 months after starting, then every 6-12 months.

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References

Accord. (n.d.). *Naltrexone* [Photograph]. Accord. https://www.accordhealthcare.us/product/naltrexone-hydrochloride-tablets/?variation_id=1505

My Health Explained. (n.d.). *Metformin* [Photograph]. My Health Explained. <https://www.myhealthexplained.com/diabetes-information/diabetes-articles/benefits-of-metformin-a-wonder-drug>

Well Rx Pharmacy. (n.d.). *Phentermine* [Photograph]. Well Rx Pharmacy. <https://wellrxpharma.com/product/phentermine-37-5mg/>